

Los Angeles General Medical Center

Creating a Patient-Centric Wayfinding System

Fall 2025 Final Deliverable

International Consulting Club (GLOBE)
at the University of Southern California

Executive Summary

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Key Question

How can LA General Medical Center redesign its wayfinding system to make navigation more **intuitive, inclusive, and efficient** for patients and visitors across its campus?

Challenges

Operations

- Complex multi-building campus
- Outdated signage and directories
- High staff reliance for navigation

Design

- Inconsistent naming conventions
- English-only, text-heavy signage
- Underutilized hybrid tools

Strategies

1. **Effective Pre-Arrival Patient Communication:** Multilingual, visual-driven, omnichannel communication
2. **Consistent Directional Signage:** Effective designs and placement while considering ADA-compliance
3. **Hybrid Wayfinding System:** Physical and digital maps and signage to support the entire patient journey

Evaluative Metrics

- Patient Navigation Efficiency
- Accessibility & Compliance
- Consistency & Scalability
- Financial Considerations

Expected Outcomes

- A cohesive, multilingual, and ADA-compliant wayfinding system
- Reduced navigation-related stress and confusion
- Improved clinic punctuality and overall patient satisfaction

ICC GLOBE's Key Objectives and Process Based on Project Asks

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Project Asks

1. What are the **key pain points** across the patient journey and which ones most affect patient timeliness, satisfaction, and stress?
2. How can LA General establish a **scalable model** for continuous **wayfinding improvement and cross-department coordination**?
3. In what ways can **multilingual and ADA-compliant** design principles be integrated into existing systems?

ICC GLOBE's Objective

Improve **current wayfinding framework** and existing signage

Emphasize **scalable solutions** amidst renovations

Prioritize **low-cost, high-impact** solutions over structural changes

Design **long-term protocols** for continuous refinement

ICC GLOBE's Process

Initial Research

Conduct background research to identify **key issues, analyze patient demographics, and define core patient segments** (e.g., non-English speakers, first-time visitors)

On-Site Assessment

Perform **campus walkthrough** to document navigation behaviors, signage inconsistencies, and accessibility gaps across major buildings and entrances

Case Studies

Analyze **case studies** from hospitals and large-scale public infrastructures (e.g., airports, metro and bus systems, stadiums) to identify **transferable best practices**

Actionable Steps

Design **sample signage and marketing material** based on evaluation of current designs, consulting architecture professionals, and additional secondary research



Patient Journey

Addressing pain points from pre-arrival to appointment

Patient Journey

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Patient Journey

	Pre-Arrival	Arrival	Outdoor Wayfinding	Indoor Wayfinding
Areas of Improvement	• Appointment emails only has room details • Lack image-driven, visual instructions • Outdated information causing confusion with Keck Hospital	• Parking assignment and location unclear • Valet services unused and underpublicized • Public transit and shuttle riders lack clear drop-off instructions	• Existing lamp-post signs are hard to read • Building labels inconsistent or missing • No indication of connected hallways between buildings	• Outdated maps and inconsistent floor numbering • Digital directories show conflicting information • Heavily reliant on staff to give directions
Action Steps	Redesign appointment confirmations/ pre-arrival information emails Send via multiple communication platforms with different languages	Promote valet service Create clear parking signage Update information found on 3rd party maps	Place lamppost and board signage at high-traffic intersections Design legible, clear, and high-contrast board signage	Update digital displays for simple navigation Establish color cues in long hallways Create Indoor Map by working with Google

Patient Journey: Pre-Arrival

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Patient Journey

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Redesigning Pre-Arrival Information

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Problem	Solution	Justification
Text-heavy information makes patients reluctant to read	• Relay information in bullet form • Highlight section headers	Allows patients to identify important information more efficiently
Navigation instructions are difficult to understand	• Attach photos of entrances • Add short videos of pathways	Enables patients to follow along the path using pictures and videos
Difficult for first-time visitors/ patients to decode	• Replace codes with plain-language names and symbols	Can easily ask staff for help and easier to interpret for non-English speakers
May miss or ignore the email if they are not active	• Automate multiple emails • Reach out via other platforms	Reminds patients of their appointment and reiterates instructions

Email Template

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1 Link Sections for Easy Navigation

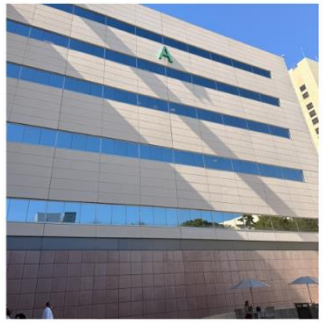
Subject: Important Information for Your Upcoming Visit to LA General
Hello **[Patient Name]**,
We look forward to seeing you at **Los Angeles General Medical Center** for your appointment on:
📆 **Date:** [Insert Date]
🕒 **Time:** [Insert Time]
📍 **Clinic/Department:** [Insert Clinic Name]
🏢 **Building:** [Insert Building Letter or Name: A, C, H, etc.]
To help make your visit smooth, please review the step-by-step guidance below including the correct parking area and the easiest route to your clinic.

1. Things to Bring
2. Parking & Getting to the Correct Building
3. Getting to Your Clinic
4. Your Visit Process
5. Contact Information

2 State Specific Locations + Promote Valet

🚗 **Parking & Getting to the Correct Building**
We kindly ask you to park in the **Parking Structure 5** as it is closest to Building A where your appointment is at. After parking, follow signs for Main Entrance / Building A Lobby.

We offer **free valet service** if you find it more convenient! Meet us at the green "Valet Parking" sign in front of Building A!



3 Specific Instructions and Labels

🚶 **How to Navigate Once You Arrive**
LA General is a large campus, so here is the clearest route to follow to get to your appointment at **2P61 Women's Health Clinic**:

1. Enter through the Main Lobby (Building A).
If you parked in Structure 5, the walkway leads directly to the Main Entrance.
2. Follow the building letter signs at major intersections.
- Signs marked **A**, **C**, and **H** indicate each building's main hallway
- Large directional arrows indicate turns at key decision points

3. Once in the building, take an elevator to the 2nd floor
4. Follow the Dark Blue line labeled 2P to the Women's Health Clinic

If you need assistance, any staff member can guide you.

4 Map Out Entire Journey

🩺 **Your Visit: What to Expect**
1. Pre-Visit Tests (if needed)
Depending on your appointment, you may complete:
- Blood work
- Vital signs screening
- Imaging or diagnostics

Building A → A2 or A3 typically handles labs and imaging.

2. Clinic Appointment
Your provider will meet with you to review symptoms, discuss treatment, and plan next steps.

3. Pharmacy Pickup (if prescribed)
The main outpatient pharmacy is located in **Building H (H1)**.

5 Contact Information

📞 **Need Help Before Your Appointment?**
If you need help determining:
- Which building your appointment is in
- The best place to park
- Mobility or disability accommodations
- Directions on arrival

Please call **[Clinic Phone Number]**.

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Additional Platforms to Convey Pre-Arrival Information

The Problem

Information is not centralized or accessible across different communication touchpoints

Pre-arrival information is buried in long text emails and easily overlooked

Strengthen pre-arrival communication through multiple channels to ensure patients arrive at the right building on time

Recommendation

Platform	Function	Benefit
Email	Send appointment details with building, floor, and parking info	Embed visuals and interactive campus map
SMS / WhatsApp	Deliver day-before and day-of reminders	Direct map links for real-time navigation
Patient Portal (MyWell / MyChart)	Appointment confirmation and directions to building	Store pre-arrival info and videos
LA General Website	Centralized digital hub for visit preparation	Upload short “How to Get Here” videos and FAQs by department

Patient Journey: Arrival

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Valet and Parking Service



Promotion (Inclusion of Valet)

- Include valet and parking details in appointment confirmations and patient information
- Display valet parking icon on the map (P5) and signage
- Include a slide on valet parking in the on-site directories to emphasize valet is free

Patient Experience

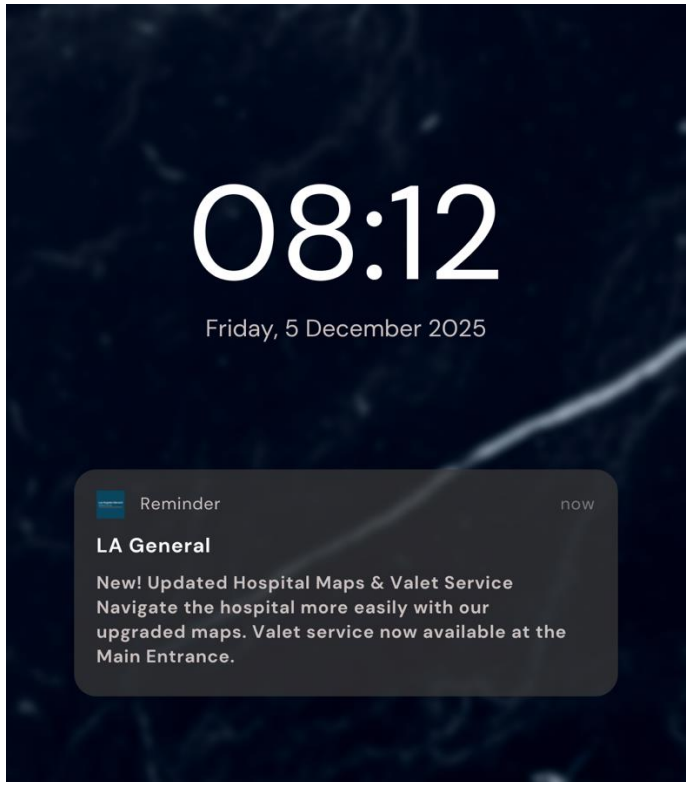
1. **Arrival:** Patient drives to P5
2. **Signage:** Clear valet parking signs and arrows guide them to drop-off area
3. **Valet Assistance:** Staff assist with parking and provide a ticket
4. **Wayfinding:** Directional signage from Lot 5 to destinations (e.g. Building A, B, C, H)
5. **Return:** Patient presents a valet ticket

Valet and Parking Service Marketing

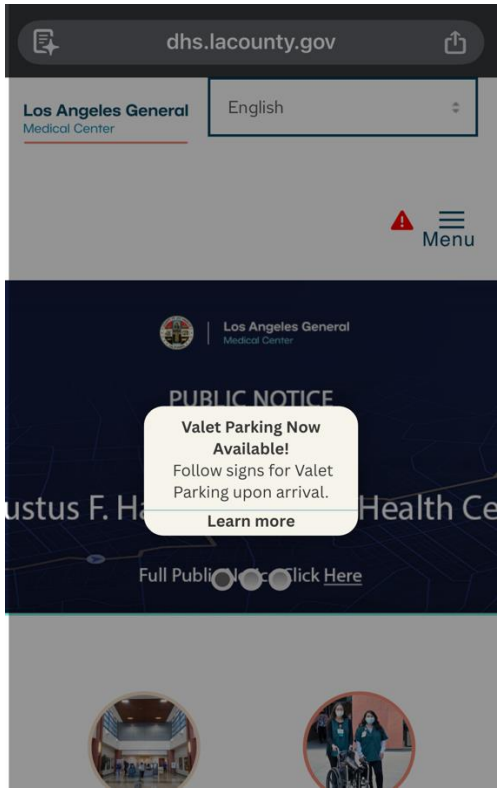
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3 Main Channels to Promote Valet and Parking Services

Appointment Confirmation Message



Banner on Main Website



Updated Phone Call Script

"Happy to help. Reminder that if you're driving here, there is valet parking available at x place. If you don't see it right away, there are signs outside that will guide you to the valet drop-off."

Improving Navigation from Parking to Buildings A–B–C–H

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Challenges

Inaccurate or outdated mapping leads to wrong building or parking arrivals

- Many visitors rely on **Google Maps or Apple Maps** rather than hospital signage
- Building names (A / B / C / H) and parking zones often appear inconsistently or not at all
- Map providers depend on **verified business listings and user reports**, not internal facilities data

Proposed Solutions

Verify & Standardize Internal Data

- Confirm exact building names, coordinates, and visitor entrances with Facilities
- Create a master location database including Official names (Building A – Ambulatory Services, etc.), Street addresses or latitude/longitude, and Preferred parking lot & accessible routes

Submit & Maintain Verified Listings

- Register each building and parking lot through **Google Business Profile Manager** → "Add a place" or "Edit location," and **Apple Business Register (Apple Maps Connect)**
- Upload the verified data, photos, and working hours; assign **hospital webmaster or communications lead** to maintain quarterly reviews

Impact

Accurate Directions

Efficient Parking

Feedback Cycle

Patient Journey: Outdoor

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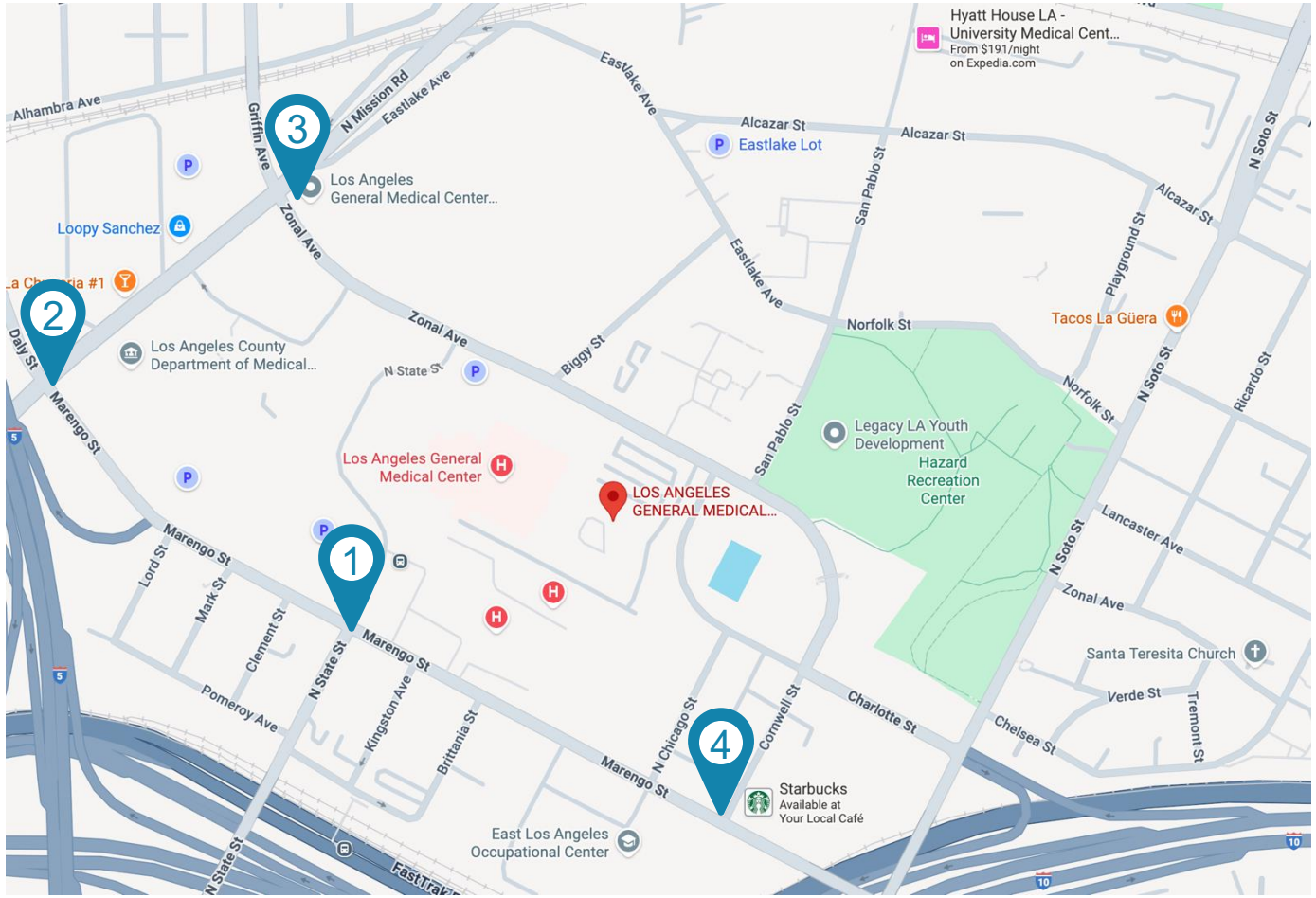
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Placement of Outdoor Directional Signs

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Optimal Locations for Directional Signs



1. **State St & Marengo St**
   Primary vehicular entry

2. **Marengo St & Mission Rd**
   Pedestrian-heavy, near bus stop for commuters

3. **Mission Rd & Zonal Ave**
   Near bus stop, large crosswalk

4. **Marengo St & Cornwell St**
   Most confused area on the LA General map

Parking Lot Entrances / Exits
→ P1 & P2 Garage exits
→ Surface lots along State St & Zonal Ave
→ Elevator lobbies must also have duplicated signs

Pedestrian Crosswalks & Paths Connecting Buildings

Design of Outdoor Signs

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Recommended Fonts for Signage Design

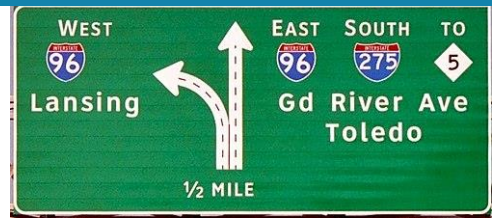
Frutiger

- → Used by: Japan JR East and West Metro, Bay Area Rapid Transit, Amtrak, BC Ferries
- → Readable from a distance



Clearview

- → Used by: U.S. Highway signs
- → Maximizes legibility at speed and distance
- → Great for outdoor wayfinding poles and parking direction



Helvetica Neue/ Helvetica

- → Used by: Washington Metro, LAX, Vancouver Airport (YVR), NYC Subway
- → Clean, professional, high contrast



Placement Recommendations

Building letters should appear at **human-eye level** (4–7 feet)

Repetition of signs **every 20–30 feet** around the building perimeter

Add markers **every 50–70 feet** from main roads, drop offs, and pedestrians to hospital parking & entrance

Place a letter marker at:

- Every main entrance
- Every secondary entrance
- Sidewalk corners

Ensure signages are **updated** whenever there are geographical changes within and surrounding the hospital

Hospital Navigation Through Effective Signage Design

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Signage Design Guidelines to Improve Visibility and Readability

Readability Rule: 1 inch letter height = ~10ft clear viewing distance

Sign Type	Letter Height
Overhead Directional Sign	2.5 ~ 4 inch
Wall Directional Sign	1 ~ 2 inch
Information Desk	0.5 ~ 1.5 inch
Door / Room ID Sign	0.5 ~ 1.5 inch

Category	Background/Text Color
Main Wayfinding	Yellow - Black
Emergency / Critical Areas	White - Red
Exits / Safety Routes	Green - White
Departments & Room ID Signs	White - Blue

Patient Journey: Indoor Wayfinding

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Standardizing and Simplifying Digital Directories to Improve Accessibility and Readability

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Recap: Challenges include information overload, limited language availability, and inconsistencies.
Suggested solutions included reformatting, other languages accessible via QR code, and standardization.

Suggested Changes



Removing Alphabetization

Switching from alphabetization to ordering departments by location (ex. A2A, A2B, A2C)



Central Alignment

Aligning department names on the left and department locations on the right



Editing the Header and Footer

Putting more emphasis on the word “Directory” instead of “Welcome” and removing the footer entirely

Effects

More Intuitive Visual Navigation

Provides easier visual navigation that allows visitors to find the location first and double check the department name

Creates Cleaner Lines

Creates a more aesthetic and cohesive appearance by creating lines that are easy for the eye to follow

Allows Space for QR Codes

Moving around and removing these elements creates space to reorganize and paste in QR codes with additional languages

Standardizing and Simplifying Digital Directories to Improve Accessibility and Readability

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Added pre-fixing for each floor for faster navigation

Switched to central alignment to create cleaner lines

Established font-size hierarchy

Ordered numerically by location for more intuitive system

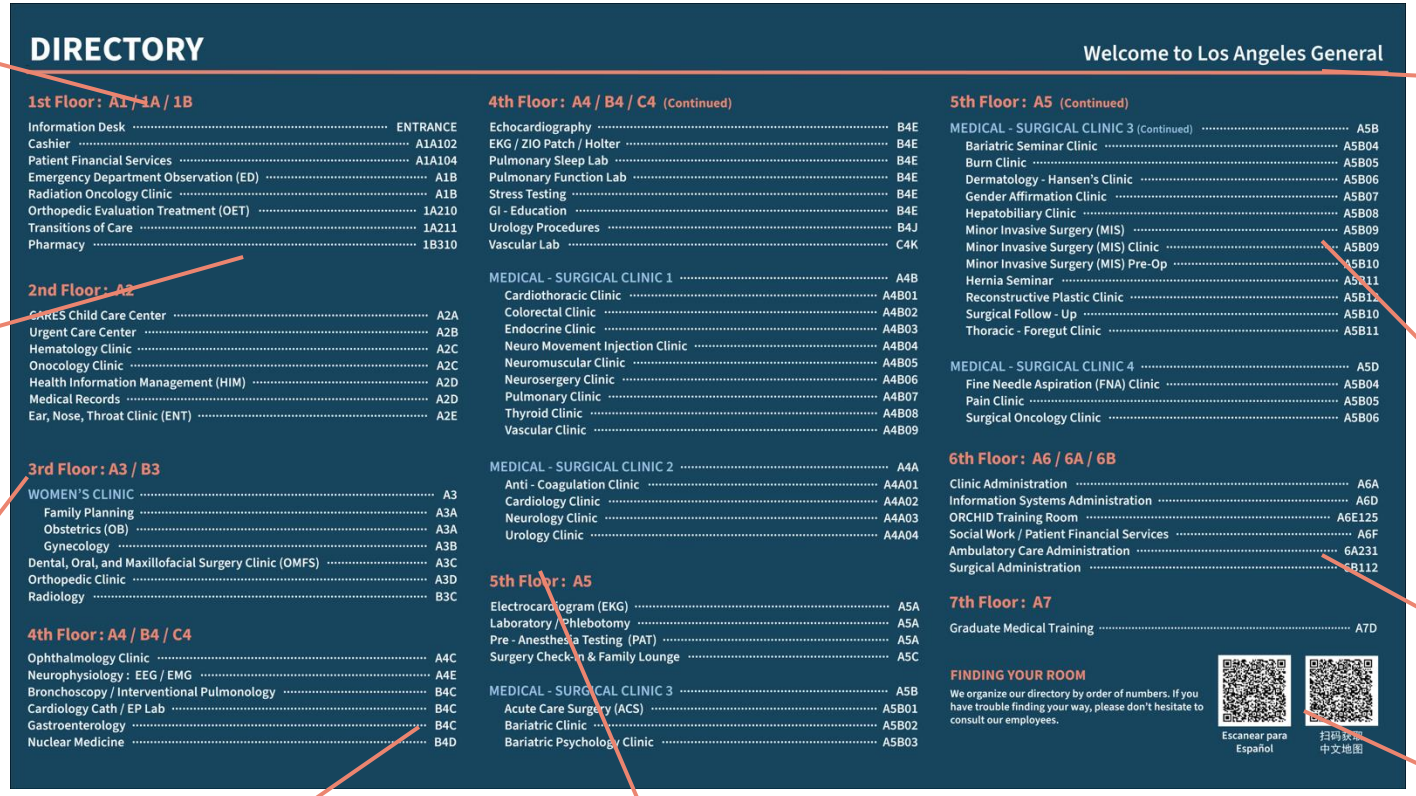
Added space between hyphens and colons

Removed footer and bottom line to create space for navigation note and QR codes

Adjusted margins to better format information

Introduced sub-numbering for Medical-Surgical Clinics

Updated header to emphasize “Directory”



DIRECTORY

Welcome to Los Angeles General

1st Floor: A1 / 1A / 1B

Information Desk ..... ENTRANCE
Cashier A1A102
Patient Financial Services A1A104
Emergency Department Observation (ED) A1B
Radiation Oncology Clinic A1B
Orthopedic Evaluation Treatment (OET) 1A210
Transitions of Care 1A211
Pharmacy 1B310

2nd Floor: A2

CARES Child Care Center A2A
Urgent Care Center A2B
Hematology Clinic A2C
Oncology Clinic A2C
Health Information Management (HIM) A2D
Medical Records A2D
Ear, Nose, Throat Clinic (ENT) A2E

3rd Floor: A3 / B3

WOMEN'S CLINIC A3
Family Planning A3A
Obstetrics (OB) A3A
Gynecology A3B
Dental, Oral, and Maxillofacial Surgery Clinic (OMFS) A3C
Orthopedic Clinic A3D
Radiology B3C

4th Floor: A4 / B4 / C4

Ophthalmology Clinic A4C
Neurophysiology: EEG / EMG A4E
Bronchoscopy / Interventional Pulmonology B4C
Cardiology Cath / EP Lab B4C
Gastroenterology B4C
Nuclear Medicine B4D

4th Floor: A4 / B4 / C4 (Continued)

Echocardiography B4E
EKG / ZIO Patch / Holter B4E
Pulmonary Sleep Lab B4E
Pulmonary Function Lab B4E
Stress Testing B4E
GI - Education B4E
Urology Procedures B4J
Vascular Lab C4K

MEDICAL - SURGICAL CLINIC 1 ..... A4B
Cardiothoracic Clinic A4B01
Colorectal Clinic A4B02
Endocrine Clinic A4B03
Neuro Movement Injection Clinic A4B04
Neuromuscular Clinic A4B05
Neurosergery Clinic A4B06
Pulmonary Clinic A4B07
Thyroid Clinic A4B08
Vascular Clinic A4B09

MEDICAL - SURGICAL CLINIC 2 ..... A4A
Anti - Coagulation Clinic A4A01
Cardiology Clinic A4A02
Neurology Clinic A4A03
Urology Clinic A4A04

5th Floor: A5

Electrocardiogram (EKG) A5A
Laboratory / Phlebotomy A5A
Pre - Anesthesia Testing (PAT) A5A
Surgery Check-in & Family Lounge A5C

MEDICAL - SURGICAL CLINIC 3 ..... A5B
Acute Care Surgery (ACS) A5B01
Bariatric Clinic A5B02
Bariatric Psychology Clinic A5B03

5th Floor: A5 (Continued)

MEDICAL - SURGICAL CLINIC 3 (Continued) A5B
Bariatric Seminar Clinic A5B04
Burn Clinic A5B05
Dermatology - Hansen's Clinic A5B06
Gender Affirmation Clinic A5B07
Hepatobiliary Clinic A5B08
Minor Invasive Surgery (MIS) A5B09
Minor Invasive Surgery (MIS) Clinic A5B09
Minor Invasive Surgery (MIS) Pre-Op A5B10
Hernia Seminar A5B11
Reconstructive Plastic Clinic A5B12
Surgical Follow - Up A5B10
Thoracic - Foregut Clinic A5B11

MEDICAL - SURGICAL CLINIC 4 ..... A5D
Fine Needle Aspiration (FNA) Clinic A5B04
Pain Clinic A5B05
Surgical Oncology Clinic A5B06

6th Floor: A6 / 6A / 6B

Clinic Administration A6A
Information Systems Administration A6D
ORCHID Training Room A6E125
Social Work / Patient Financial Services A6F
Ambulatory Care Administration 6A231
Surgical Administration 6B112

7th Floor: A7

Graduate Medical Training A7D

FINDING YOUR ROOM

We organize our directory by order of numbers. If you have trouble finding your way, please don't hesitate to consult our employees.

Escanear para Español

扫码获取中文地图

Evaluating Color Cues For Indoor Wayfinding in the General Hospital



Effective

- ✓ Color and design coding to distinguish pathways
- ✓ English and Spanish translations appeal to the diversity of patients at the hospital



Ineffective

- x Thin lines with no indication of direction
- x Similar tones: 3 tones of brown make it difficult to distinguish the line

Using Color Cues For Indoor Wayfinding

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Implementation Considerations





Tokyo Metro System

- Start point pathway identification with destination, **English and Spanish** translation
- **Indication of direction** to follow using arrows on the lines
- **Logos** on the line to identify the destination
- **Bright and thick lines** that are easy to distinguish

Color Cues Visualized

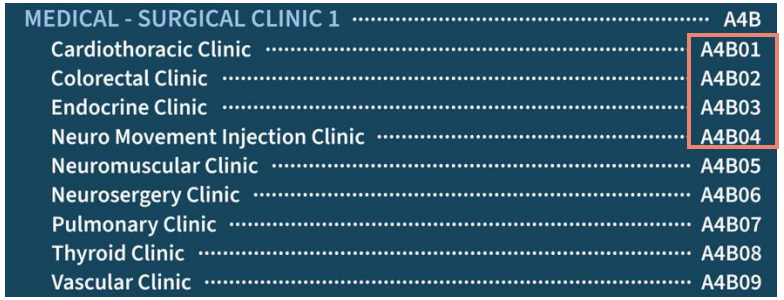


Using Color Cues For Indoor Wayfinding

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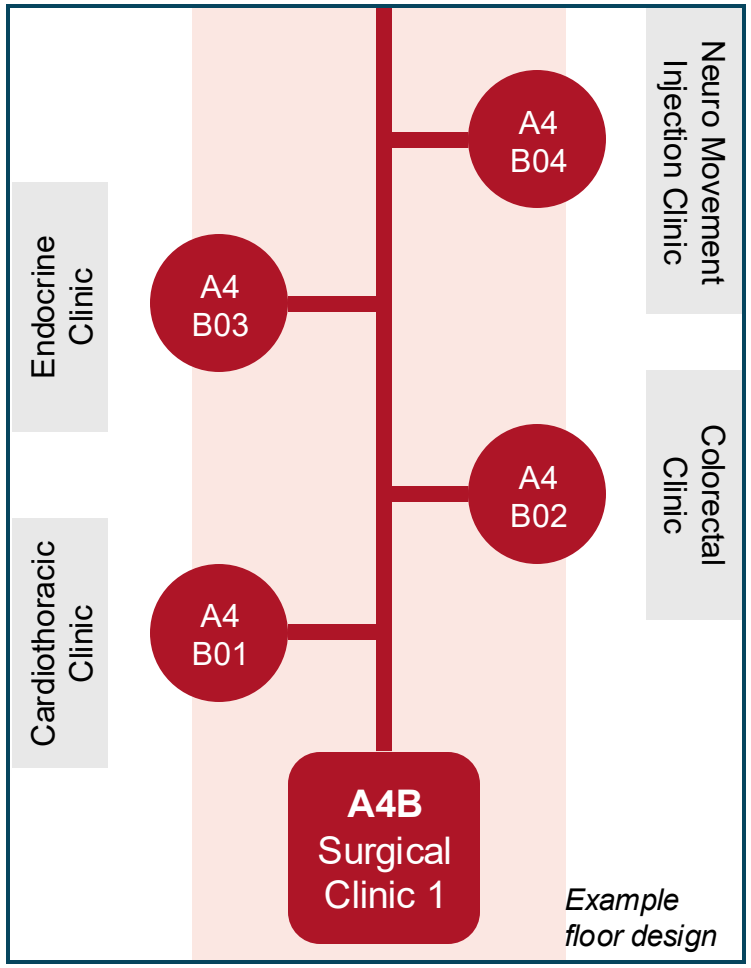
Color Cues Visualized: Example for Building H, Fourth Floor

Example Hallway



Integrated Navigation

1. Patients can use digital display, hallway signs, and pre-appointment email to help them interpret color cues
2. Numbered stopping points will be indicated for patients to navigate long hallways



On the Floor…



On the Wall/ Door…

A4B02
Colorectal Clinic
Clínica Colorrectal

Rationale for Color Codes

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Colors	Reason
Green Lobby, Waiting areas, Food court, Gift shop, Place of worship, Parking	Calm and Welcoming
Red Emergency, Surgery, Cardiology, Intensive Care, Infectious diseases	Vibrant and Urgent
Blue Obstetrics, Maternity, Pediatrics, Family medicine	Stable and Relaxing
Soft Blue Dermatology, Plastic surgery, Orthopedics, Physical therapy, Prosthetics	Reassuring and Safe
Purple Oncology, Cancer care, Radiology, Rheumatology, Genetics, Mental health, Spiritual care	Sympathetic and Reserved
Yellow Pediatrics, Family Medicine, Social services	Joyful and Buoyant
Orange Medicinal library, Medical supplies, Medical records, Conference rooms	Vivid and Energetic
Grey Staff only, Utility areas, Mechanical areas	Neutral and Professional

Color Guide Pamphlets

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Color Code

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- Emergency, Surgery, Cardiology, Intensive Care, Infectious diseases
- Medicinal library, Medical supplies, Medical records, Conference rooms
- Pediatrics, Family Medicine, Social services
- Lobby, Waiting areas, Food court, Gift shop, Place of worship, Parking
- Dermatology, Plastic surgery, Orthopedics, Physical therapy, Prosthetics
- Obstetrics, Maternity, Pediatrics, Family medicine
- Oncology, Cancer care, Radiology, Rheumatology, Genetics, Mental health, Spiritual care
- Staff only, Utility areas, Mechanical areas

Access Via 3 Main Channels



QR Codes

- Located at major decision points
- Cost Effective
- Multilingual
- Sound features for visually impaired



Display Screens

- Added amongst other updates
- QR codes
- Large and visible
- Screens already in use



Paper Copies

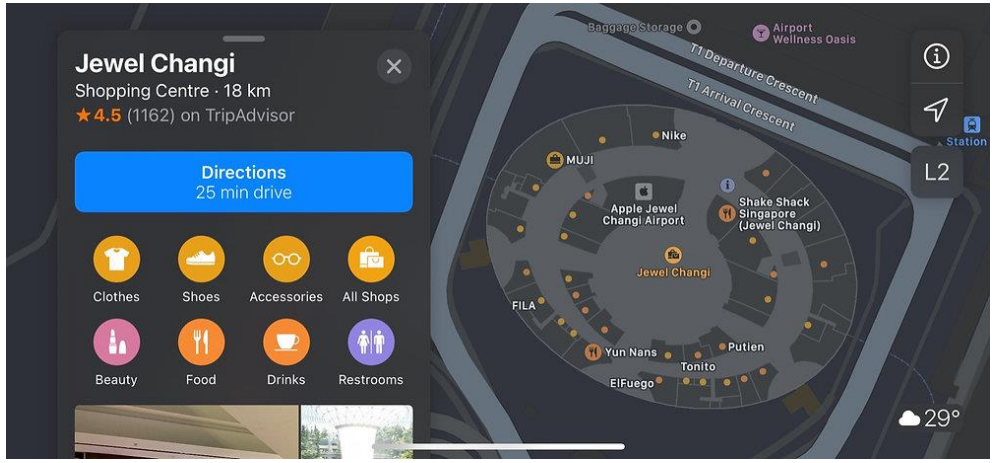
- Provided with map upon entry
- Stored on each floor
- Incurs cost over time
- Limited copies and translations

Partnership with Google Indoor App

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Case Study: Singapore Changi Airport



Key Features

- Real-time **blue-dot indoor navigation** using Wi-Fi and Bluetooth beacons
- **Turn-by-turn directions** across terminals, shops, rest areas, and gates
- **Multilingual support** for international travelers
- **Integration** with different retail stores and transport links that lead to the airport

Takeaways and Implications

Link indoor navigation to **appointment confirmations** (via Google Calendar, patient portals)

Leverage Google's **multilingual and ADA-compliant features** (e.g, voice guidance)

Minimizing construction costs amidst construction and renovations in the coming years

Impact

- Reduced confusion and congestion
- Improved accessibility for users with disabilities
- Enhanced user experience through accurate, dynamic, and visually intuitive maps

Considerations

- Requires staff training and continuous IT support for updates and maintenance
- Dependence on stable Wi-Fi or Bluetooth
- Tech literacy of older population may be an issue

Google Indoor App Application Process

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Application

- To have an indoor view on Google Maps, facilities and institutions must already be a Google Maps Indoor Partner
- To become a partner, institutions must contact the appropriate regional Indoor Partner group (indoorpartners-americas@google.com)

Material

- After acquiring partnership, they must submit the appropriate materials developers to generate an indoor view within Google Maps:
  - Updated floorplan for each facility
  - Indications of important wayfinding services (movement services, information/reception desks) and other POIs (department clusters)
  - Physical scanning for Indoor Navigation API

Integration Timeline

Contacting developers and becoming google indoor partner (~1 week)

Gathering and submitting necessary materials (~1 month)

Indoor API Development (~3-6 months depending on scale)

Ensuring Patient Access and Use



QR codes posted in and around buildings



Hyperlink on website



Staff References

Google Indoor App Pricing Plan

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LA General needs to pay per API (Application Programming Interface) call under the Google Maps Platform pricing

API Pricing = Basic Pricing List + Possible Subscription Bundles

Basic Pricing List

- Categories: Maps (Dynamic / Static, Elevations), Routes, and Places (Geocoding, Address Validation, Time Zones)
- For each SKU, business needed to pay a price ranging from $5 to $40 per 1000 mapload
- Ex: Compute Routes Essentials cost $5 per 1000 mapload while Roads Speed Limits costs $20 per 1000 mapload

Possible Subscription Bundles

- Categories: Standard, Premium, Enterprise
- Recommendation for LA General: Enterprise Plan
- Offer customized pricing based on specific requirements and extensive mapping needs of large businesses, ensuring large organizations to manage cost effectively

Possible Additional Fees

Floor Plan Preparation Costs

Cost: $2000 to $10000
Clear and standardize the existing floor plans, changing them into indoor-mapping format.

Professional 3D Scanning

Cost: $10000 to $25000+
Physical scanning of indoor floorplan helps to build accurate capture of corridors, rooms, and waiting areas.

Ongoing Maintenance Costs

Cost: $1000 to $5000 per year in labor and $2500 to $20000 per year on vendor maintenance.
Updating plans and pathways

Google Indoor App Alternatives

Google Indoor App Alternatives

Ranking	Platform	Main Features	Cost	Trade-off
1	Mapsted	Indoor/outdoor maps, built-in blue-dot (no hardware), multi-floor/building support, accessible across mobile/web/kiosk	Low hundreds $/mo for mid-size venues	Slightly less precise than hardware-based systems
2	Mappedin	2D/3D interactive maps, web/mobile/kiosk embed, live map editing; optional blue-dot with IPS hardware	$165–200/mo per map (basic), plus hardware if blue-dot	Blue-dot needs extra hardware or IPS setup; ongoing maintenance
3	22Miles	2D/3D maps, multi-floor/building wayfinding (include ADA routes), kiosks, digital signage, QR-to-phone	$10K–50K/yr depending on # of kiosks/screens	Higher upfront cost, infrastructure-heavy
4	Pointr	High-accuracy blue-dot navigation, analytics, AI-assisted maps, SDK/API integration	$10K–50K+/yr	Expensive, complex



Summary

Summarize our recommendations and looking into next steps

Executive Summary

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Key Question

How can LA General Medical Center redesign its wayfinding system to make navigation more **intuitive, inclusive, and efficient** for patients and visitors across its campus?

Challenges

Operations

- Complex multi-building campus
- Outdated signage and directories
- High staff reliance for navigation

Design

- Inconsistent naming conventions
- English-only, text-heavy signage
- Underutilized hybrid tools

Strategies

1. **Effective Pre-Arrival Patient Communication:** Multilingual, visual-driven, omnichannel communication
2. **Consistent Directional Signage:** Effective designs and placement while considering ADA-compliance
3. **Hybrid Wayfinding System:** Physical and digital maps and signage to support the entire patient journey

Evaluative Metrics

- Patient Navigation Efficiency
- Accessibility & Compliance
- Consistency & Scalability
- Financial Considerations

Expected Outcomes

- A cohesive, multilingual, and ADA-compliant wayfinding system
- Reduced navigation-related stress and confusion
- Improved clinic punctuality and overall patient satisfaction



Appendix

Additional research, alternative recommendations, and supporting material

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Patient Journey

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Supporting Material

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S2: Directional Sign Samples
S3: Digital Display Design

A1: Training Staff for Facility Familiarity

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Integrating Wayfinding into Staff Onboarding

Facility-Wide Training



- Digital facilities orientation (video guide, interactive map)
- **Focus**: Patient entrances/exits, receptions desks, department clusters, critical patient-facing routes

Departmental Training



- Short in-person briefing/tour
- **Focus**: Local layout and nearest kiosks, adjacent departments, and exit routes

Ensuring Ongoing Consistency of Information

Facility Updates



- Utilize current modes of communication: email, staff newsletter, departmental briefings
- **Focus**: Local room changes, department shifts

Staff Re-training



- Renew digital orientation annually
- **Focus**: Changes to facility layout, new information kiosks/ desks

A2: Updating Maps In the Long-Term

Recommended Actions for Physical and Digital Maps

Wayfinding Team

- A dedicated wayfinding team to ensure all old maps are taken down or collected
- Pair the map update check with the equipment safety inspection process (3 ~ 6 months)

Map Design

- Add date on the bottom of the map of when the map was distributed

Website

- Upload newest version of the campus map and floor map of each building on LA General Medical Center website
- Write a disclaimer regarding when the newest map change was and brief bullets describing the change made

Google Maps

- Make sure that each building is labeled correctly on Google Map
- Give specific instructions for patients after they scheduled an appointment (what destination name/address they should input)

Case Study: New York Subway

Problem

- A newly opened station could take 12–18+ months to appear on the station map boards
- Service changes (especially due to construction) were handwritten taped-on printouts

Proposed Solutions

- The wall map was physically gridded into 6–12 rectangles
- Divided NYC into “hot zones” vs “cool zones” (hot zones change more frequently).
- Station agent “trigger report”

Impact

- Lower printing waste
- More up-to-date
- Less customer complaints

Appendix

A3: Improving Navigation from Parking to Buildings A–B–C–H: Integrated Physical and Digital Wayfinding

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Physical Infrastructure Enhancements

Flat, large outdoor directory panels

- **3D site map** (simple axonometric, color-coded)
- **"You Are Here" marker** + **estimated walking times**
- High-contrast matte surface, bilingual
- *Optional:* QR code linking to live navigation app (see right section)

Pathway Signage & Landmarks

- Color-coded directional pylons every 150 ft
- Add **tactile arrows** and **non-slip contrast strips** along accessible routes
- Lighted overhead signs at building entrances for nighttime visibility

Environmental Cues

- Paint **ground route lines** from parking lot zones to building entries
- Use consistent naming and icons across maps, pylons, and doors

Digital Wayfinding Integration

Outdoor Kiosks

- Touch-enabled screens co-located with physical directories
- Offer **step-by-step route visualization**, accessibility toggle (wheelchair, low-vision), and multilingual options

Mobile Navigation Layer

- Integrate site map into **hospital mobile app** and **QR codes** placed on all physical signs.
- Provide **turn-by-turn walking guidance** between A–B–C–H using GPS + Wi-Fi triangulation.

Real-time Information Dashboard

- Displays on app and kiosks: **parking availability, building entry points, and route obstructions** (e.g., maintenance)
- Can be accessed through mobile devices to reduce cost of extra kiosks

A4: Installing Kiosks to Improve Patients' Ability to Navigate Individually

Leveraging Kiosks for Indoor Navigation

Implementation of Kiosks

- Kiosks can help patients identify specific locations without relying on human support at the information desk
- Placed at the information desks after security check
- Placed on every floor in front of entrances (e.g., elevators)

Benefits of Kiosks

- Improve patients' ability to **navigate through the facilities by themselves**
- **Reduce workload** for workers to guide patients to their intended locations

Kiosk Type	Cost Range	Key Features
Simple Digital Display	$1,500–$4,000	Static touch screen, preloaded maps
Standard Interactive Touchscreen	$5,000–$10,000	Real-time navigation, multilingual, higher interactivity
Custom / Smart	$10,000+	Augmented reality, mobile sync, advanced integrations
Outdoor	~$20,000	Weather resistant, durable hardware

A5: Environmental Design Interventions & Implementations

Research shows: Hospitals implementing comprehensive wayfinding strategies combining environmental design elements achieve dramatic improvements

Color Psychology to Reduce Stress



Expand two-color departmental system (doorways, floors, walls) to simplify navigation
Use calming blues/ greens with warm accents to reduce stress

Circadian Lighting Pilot



Deploy tunable white LED systems (2700K–6500K) in long-stay units (ICU, behavioral health, oncology)
Benefits: improved sleep, mood, and recovery

65% of patients report better experiences, **20%** fewer missed appointments, and **35%** reductions in perceived wait times

A6: Comparison of Initial Setup and Ongoing Annual Costs for Internal Wayfinding Solutions Mentioned Above

Costs Comparison

Kiosks



- Initial setup: $220,000
- Additional software, installment, & maintenance: $1,000

Staff Training



- Coach & materials: $150-$300
- Lost productivity: $360,000 (9,000 staff × 2 hrs × $20/hr)

Color Cues



- 16,000 LF to be painted
- Paint cost @ $1 per LF = $16,000

Physical Maps



50-100 copies per floor (~18 floors, $1 print per copy): $1400

Digital Displays



- Software cost: ~$0
- Annual design labor: $1,000

Estimated total costs: $599,000

A7: Building an Accessibility Ecosystem, Connecting Communication, Comprehension, and Mobility into the User Experience…

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Hearing Impairments

Missing visual equivalents for auditory information

- Uncaptioned displays, no visible interpreter signage, **inconsistent** hearing-assist symbols
- **ADA §703.7**: hospitals must ensure that communication with people with hearing disabilities is *"**as effective as** that provided to others"*

Cognitive Impairments

Wayfinding becomes a psychological challenge

- ~ 40% elderly hospital patients experience **cognitive disturbance** or disorientation during visits
- For cognitively impaired patients, **long, jargon-filled instructions** or inconsistent department naming increase **anxiety** and delay care

Mobility Impairments

Different line of sight often overlooked

- Blocked routes, improper signage heights, and **glare-heavy** materials that make tactile signs unreadable
- **ADA §703.4**: tactile characters must be mounted between *48–60 inches* with ***non-glare finishes*** and *18×18 inches clear floor space*

Communication Equity

Achieve communication parity by ensuring spoken information is always available in **visual form**

Wayfinding Simplification

Design for intuitive comprehension by **minimizing decision points** for neurodiverse and elderly patients

Physical Independence

Create environments that support self-navigation with dual **Digital-Physical redundancy** in accessible paths

Unified Framework

Establish a **cross-departmental framework**; integrate ADA 2010 §703 and California Title 24 standards

Source: California Department of Rehabilitation. "Title 24 Building Standards Code and Accessibility Requirements."

A8: … Will Transform LA General's Navigation System into a Model of Sensory-Inclusive, Low-cognitive-load Design

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Recommended Actions

Hearing Accessibility

- Standardize **hearing-assist icons**, captioned displays, and **visible interpreter stations** across all high-traffic areas
- **Goal:** 100% of check-in points should support **visual + audio redundancy**

Cognitive Accessibility

- Maintain a **centralized lexicon** for signage terminology and use **ISO 7001 icons,** coordinate across departments for naming
- **Goal: 30% faster** average self-navigation by neurodiverse/ elderly patients

Mobility Accessibility

- Implement **route-protection policy** to comply with **ADA 2010 Guidelines; Require tactile / Braille compliance checks** renovation
- **Goal:** 100% of patient-facing areas are compliant, ensuring seamless navigation.

Source: United States Access Board. "Guide to the ADA Standards: Chapter 7 – Signs."

Compliance in Continued Operation

Institutionalize Accessibility

- Annual **campus audits** with photographic documentation
- **Compliance dashboard** tracking ADA §703 and Title 24 §11B-703 adherence

Staff Training and Engagement

- Develop mandatory **staff orientation modules** on inclusive communication and assistive technologies
- **Disability advocacy organization** partnerships in annual testing and audit processes

Impact: Patient-Centric Navigation

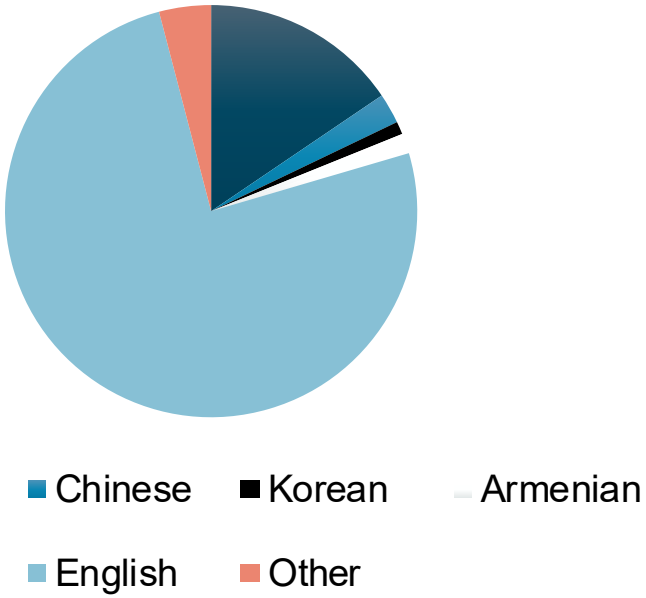
- **Reduced Wait and Search Times**
- **Better Operational Efficiency**
- **Compliance Confidence**
- **Improved Patient Experience**

A9: Addressing the Diverse Language Needs at LA General

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Diverse Linguistic Needs

Language spoken at home among the population of LA County



66.2% of patients are Hispanic/Latino, but there is a **lack of Spanish signage**

Recommendation

Create **country flag pins** for staff to wear on their chest indicating languages they speak

Spanish **translations for the main signs**, e.g., Emergency and Accessibility information

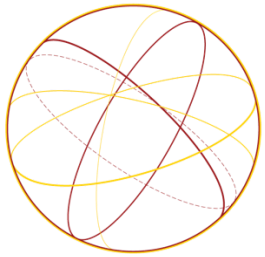
Integrate an **icon-based signage system** based on universal symbols

Place **QR codes** at the main entrance showing the **legend for icons** in the languages spoken in LA County

Impact

- Reduces anxiety and confusion for multilingual and low-literacy patients
- Improves patient flow by minimizing time lost due to navigation issues
- Fast, low-cost implementation using existing infrastructure

Sources: Government of Los Angeles County



X

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Creating a Patient-Centric Wayfinding System for LA General Medical Center

Fall 2025 Final Deliverable